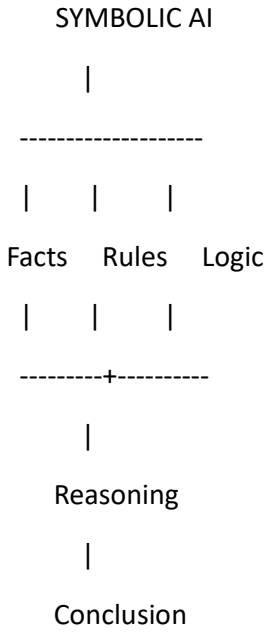
Symbolic Ai

Symbolic AI is an approach to Artificial Intelligence in which knowledge is represented using **symbols, facts, rules, and logic**, and the system uses **reasoning** to solve problems and make decisions.



Example-

If a person have -

Fever + Cough
↓
Apply Rule
↓
Possible Flu

Suppose that I give to things to computer

Fact: Ravi is a student.

Rule: If a person is a student, then the person studies.

Therefore:

Ravi is a student → Ravi studies.

Advantages

1. **Easy to understand** – Rules are clear and understandable.
2. **Explainable** – The system can explain how it reached a conclusion.

3. **Good for logical reasoning** – Useful for solving logic-based problems.
4. **No large dataset required** – Knowledge can be directly given as rules.
5. **Easy to modify** – New rules can be added or existing rules can be changed.
6. **Useful in expert systems** – Used in areas such as diagnosis, planning, and decision-making.

Disadvantages

1. **Rule creation is difficult** – Humans have to create many rules manually.
2. **Poor handling of uncertainty** – Real-world situations are often uncertain. (it may rain tomorrow).
3. **Difficult to handle complex data** – Images, speech, and unstructured data are difficult to represent.
4. **Too many rules become complex** – Large systems can have thousands of rules.
5. **Limited learning ability** – Traditional Symbolic AI does not automatically learn patterns like Machine Learning.
6. **Maintenance can be difficult** – Updating a large knowledge base can require significant effort.

Connectionist Ai

Connectionist AI is an approach to Artificial Intelligence that uses interconnected artificial neurons to learn patterns from data and make decisions or predictions.

Many Examples

↓

Neural Network

↓

Learns Patterns

↓

New Input

↓

Prediction

Another Simple Example: Face Recognition

Suppose we want an AI system to recognize **Rahul's face**.

We provide many images:

Rahul's photo 1

Rahul's photo 2

Rahul's photo 3

Rahul's photo 4

↓

Neural Network

↓

Learns facial patterns

The network learns features such as

Eyes, Nose ,Face shape ,Distance between eyes ,Other visual patterns

Example: Email Spam Detection

Suppose we want AI to identify whether an email is **Spam** or **Not Spam**.

We give the system thousands of emails.

Training data

Email	Result
"Win ₹10,000 now!"	Spam
"Congratulations! You won!"	Spam
"Meeting at 10 AM"	Not Spam
"Please send the report"	Not Spam

The neural network learns patterns from these examples.

For example, it may learn that words such as:

- "Win"
- "Prize"
- "Free"
- "Congratulations"

are often associated with spam.

Then a new email arrives:

"Congratulations! You have won a free prize."

The network processes it and may produce:

Spam probability = 98%

Not Spam probability = 2%

Therefore:

Output → SPAM

Advantages of Connectionist AI

1. Learns from examples

It can learn patterns from training data instead of requiring every rule to be manually programmed.

2. Good for pattern recognition

It is very useful for:

- Face recognition
- Speech recognition
- Image recognition
- Handwriting recognition

3. Handles complex problems

Neural networks can learn complicated relationships between inputs and outputs.

4. Improves with suitable training data

With appropriate data and training, the model can improve its predictions.

5. Useful in many applications

Used in:

- Chatbots
- Medical diagnosis
- Recommendation systems
- Self-driving technologies
- Voice assistants